

Urban Microbiome Composition and Economic Development: A Multi-Cohort Metagenomic Analysis

Research Question

Does economic development (high-income (further referenced as HIC) vs. upper-middle-income (further referenced as UMIC) countries) influence urban environmental microbiome composition, and do these differences persist after controlling for geographic confounders and study-level batch effects?

Hypothesis: The level of economic development influences the urban microbiome diversity, with higher-income cities hosting specialized microbial communities shaped by the factors of industrial pollution, sanitation and agricultural land use.

Data

Database: SPIRE Environmental Metagenomics (MetaPhlAn4 species profiles; total: 26938 samples, 391 studies; urban biome subset: 1909 samples)

Stage 1: 1417 urban samples (HIC: USA+Denmark+Singapore $n = 1247$; UMIC: China $n = 170$); 561 species after prevalence filtering; Centered Log-Ratio transformation; geographic confounders identified.

Stage 2: 135 samples with complete metadata (USA $n = 29$, China $n = 106$); 134 species; linear regression with/without geographic covariates.

Stage 3: 6 independent cohorts, $n = 1258$ (HIC 1152, UMIC 106), 104 species observed in outdoor air, indoor dust, sewage, and vehicle surfaces.

Study	Country	Income	Material	n
Gusareva 2019	Singapore	HIC	Outdoor air	388
Ohara 2017	USA	HIC	Vehicle surfaces	398
Brinch 2020	Denmark	HIC	Sewage	216
Fahimipour 2018	USA	HIC	Indoor dust	115
Qin 2020	China	UMIC	Outdoor air	106
Shaffer 2021	USA	HIC	Mixed urban	35

Table 1: Cohorts chosen for meta-analysis

Results

Stage 1: Diversity analysis. UMIC samples had significantly higher richness (71.6 vs. 51.1, $p < 10^{-6}$), Shannon (2.04 vs. 1.36, $p < 10^{-7}$), and Simpson (0.60 vs. 0.50, $p < 10^{-5}$) diversity. PCA revealed community-level separation (PC1 = 21.7%). 529/561 species identified as differentially abundant (FDR $q < 0.05$, Mann-Whitney U test): 124 UMIC-enriched, 405 HIC-enriched.

Stage 2: MWAS with covariate adjustment. Geographic variables were major confounders (longitude and elevation with $p < 10^{-3}$). Unadjusted analysis flagged 64.2% of species; after adjusting for latitude and elevation, 74.6% of species retained robust economic associations (FDR $q < 0.05$), demonstrating that economic effects are real but substantially confounded by geography.

Stage 3: Multi-cohort meta-analysis. Batch effects: Strong study-level clustering (Kruskal-Wallis $H=634$, $p < 10^{-100}$; Spearman $r_{\text{study}} = -0.53$). Ordinary least square residualization (per-species regression on study dummies; residuals retained as corrected values) reduced batch correlation to $r_{\text{study}} = 0.09$.

Pooled analysis (Centered log ratio $\sim$ Income + Study, all 6 cohorts): only 8/104 species survived FDR correction, as differences in sample material (air vs. sewage vs. dust vs. vehicle surfaces) drive stronger compositional shifts than economic group.

Air-cohort comparison (Gusareva Singapore vs. Qin China): as both studies sampled outdoor air but in different income-group countries, comparing them isolates the economic signal from the sample-type confound. Outcome: 91/104 species differentially abundant (FDR $q < 0.05$; 63 HIC-enriched, 28 UMIC-enriched).

Alpha diversity (air cohorts): UMIC Shannon 0.74 ± 0.29 vs. HIC 0.43 ± 0.47 ($p = 6.8 \times 10^{-13}$);

UMIC richness 39.8 ± 22.6 vs. HIC 2.2 ± 2.8 ($p = 3.4 \times 10^{-56}$) — measured across 104 overlap species.

Conclusions

Economic development significantly influences urban microbiome diversity across all three analyses. However, sample-type heterogeneity across independent cohorts creates strong batch effects that mask biological signal in pooled analyses. Matching by sample material (air cohorts) recovers 91 significant species, validating the covariate-adjusted results from Stage 2.



Figure 1: Figure 1. Meta-analysis. (A-B) PCA before/after OLS batch correction (colored by income group): HIC blue, UMIC orange. Before correction cohorts separate; after correction income-group signal emerges. (C) Shannon diversity per cohort. (D) Volcano plot (pooled): 2 species survive FDR correction. (E) Top 15 species by effect size. (F) Shannon diversity in air cohorts only (Singapore HIC vs China UMIC): UMIC significantly higher ($p = 6.8 \times 10^{-13}$).

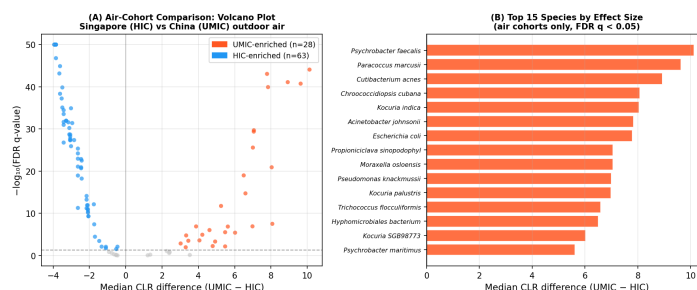


Figure 2: Figure 2. Air-cohort comparison (Gusareva 2019, Singapore vs. Qin 2020, China). (A) Volcano plot: 91/104 species differentially abundant (FDR $q < 0.05$); 63 HIC-enriched (blue), 28 UMIC-enriched (orange). (B) Top 15 species by effect size.

Comparison with Published Data

Reference: Danko et al. (2021), “A global metagenomic map of urban microbiomes and antimicrobial resistance”, Cell, (MetaSUB Consortium; 4728 samples, 60 cities).

Alignment with our findings: Both studies confirm that geography dominates urban microbiome variance over economic status, and that unadjusted cross-city comparisons are highly confounded. Both studies report globally prevalent taxa also detected in our air-cohort comparison: *Micrococcus luteus* is HIC-enriched in our data ($q = 5.7 \times 10^{-19}$), consistent with MetaSUB findings. *Cutibacterium acnes* is UMIC-enriched in our air cohort ($q = 8.0 \times 10^{-42}$), diverging from the MetaSUB pattern — likely reflecting higher human-skin aerosol load in Chinese outdoor air. The MetaSUB also found higher AMR gene prevalence in lower-income cities, consistent with the higher microbial diversity observed in UMIC environments.

Analytical differences:

1. Resistome profiling: AMR gene quantification linked income to public-health outcomes, absent

from our analysis.

2. Standardised sampling: uniform surface swabs across cities eliminated material-type heterogeneity that limited our pooled result to 8 significant species.